



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

Summary of Methodologies:

1. *Data Collection: by accessing SpaceX data via API and web scraping a Wikipedia article.*
2. *Data Wrangling: cleaning and formatting the data, feature engineering.*
3. *Exploratory Data Analysis (EDA): by creating a number of visualization plots and by SQL queries.*
4. *Interactive Visualizations: building a Plotly Dash dashboard and a Folium Interactive Map.*
5. *Predictive Analytics: training and optimizing four classification models: logistic regression, SVM, decision tree, and k-nearest neighbor.*

Summary of Results:

1. *Data Insights: discovered several factors influencing Falcon 9 first stage landings and visualized geographical and data patterns.*
2. *Model performance: all models performed well, but decision tree was the best with 94% accuracy, while the rest had 83% accuracy.*
3. *Key Findings: launch site and orbit type impact landing success, with payload mass also relevant, but a weaker factor; there was a steady improvement of success rate over the years from 2013 to 2017, then it stabilized on a high value. KSC LC-39A is the best launch site.*

Introduction

- **SpaceX** is currently the most successful of all commercial space companies. Its accomplishments include supplying and sending crews to the International Space Station, and building a satellite internet constellation Starlink.
- **SpaceX**'s workhorse, two-stage-to-orbit medium-lift launch vehicle **Falcon 9** is famous for its to-date 626 successful launches and relatively low price per launch of US\$74 million.
- A lot of the savings are due to the fact that **SpaceX** can often return and reuse the larger and quite expensive first stage of **Falcon 9**. Thus if we can predict whether **SpaceX** successfully recovers the first stage, we can estimate the cost of a launch.
- The goal of this capstone project is to study publicly available data on **Falcon 9** launches in order to reveal the patterns allowing to determine **the success rate of the first stage recovery** and **factors that influence it**.
- It can be later used to determine the cost of launches and ultimately answer the question, if another company can compete with **SpaceX**.

Section 1

Methodology

Methodology

Executive Summary: a comprehensive approach based on CRISP-DM: data collection, wrangling, EDA, interactive visual analytics, predictive analysis with classification models, and evaluation.

- **Data collection:** raw data was collected by both using SpaceX API and webscraping a Wikipedia article.
- **Data wrangling:** handled missing values, standardized data formats, extracted and engineered features.
- **Exploratory data analysis (EDA) using visualization and SQL:** visualized key metrics and parameters such as success rates, payload mass, launch sites, and orbits, their relation to each other. With SQL queries derived a number of statistical relationships in the data.
- **Interactive visual analytics using Folium and Plotly Dash:** created interactive geographical map showing the launch sites and their corresponding success rates, and a dashboard displaying success rates in relation to payload mass and launch site.
- **Predictive analysis using classification models:** 4 different classification models (Logistic Regression, SVM, Decision Tree, and kNN) were trained and the best hyperparameters selected for each of them using grid search. Their performance was compared on the same test set by generating confusion matrices and classification reports.

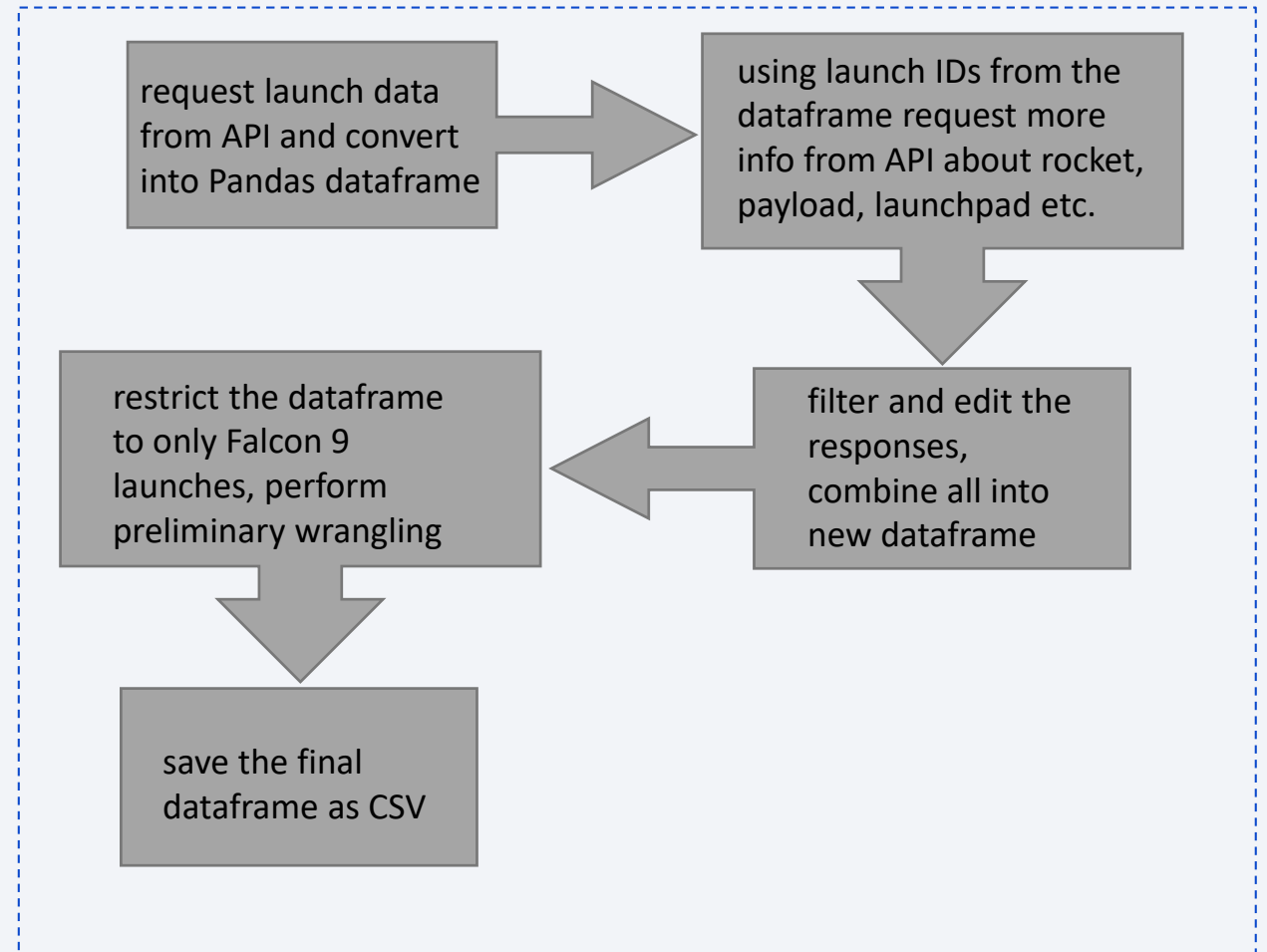
Data Collection

Datasets were collected in two ways:

- 1. through **accessing** the SpaceX **API**;*
- 2. by **webscraping** a Wikipedia page.*

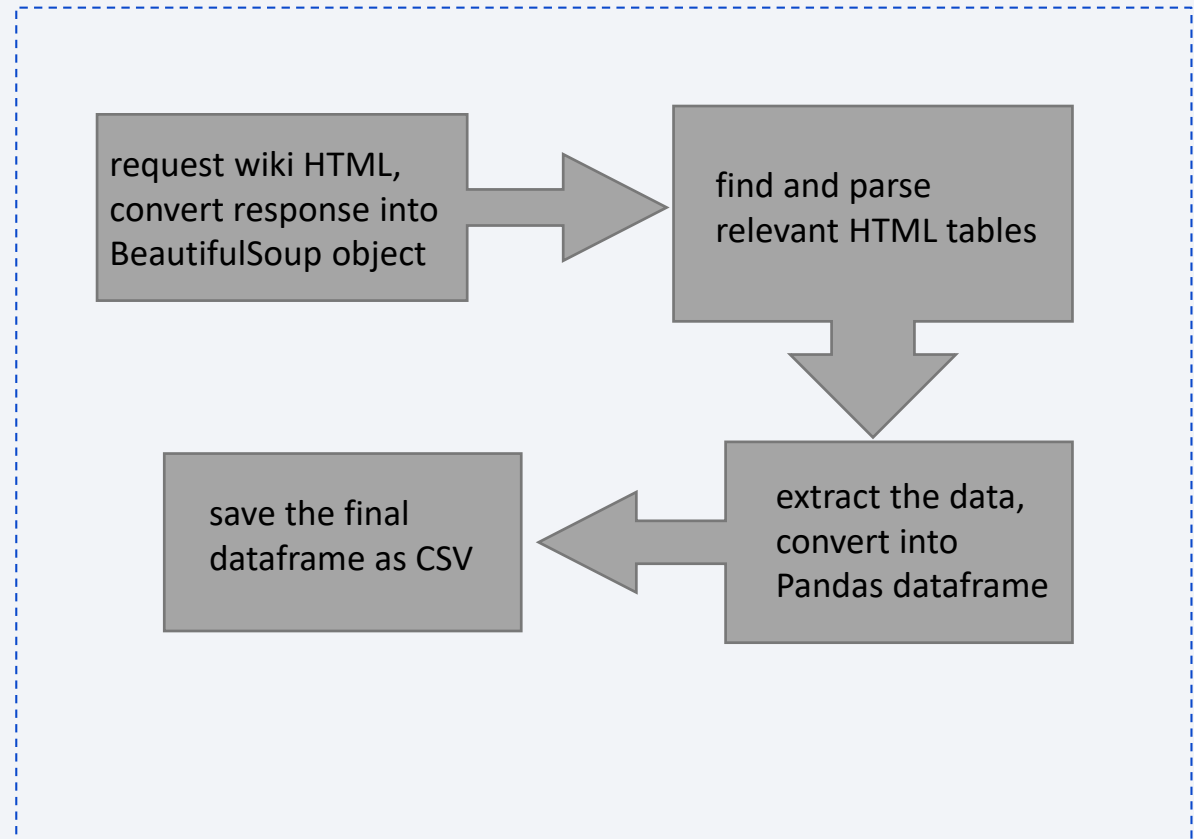
Data Collection – SpaceX API

- *Pull the relevant data from SpaceX API with multiple requests, convert to Pandas dataframe, filter and edit it, perform initial wrangling, and save as CSV file.*
- <https://github.com/yuretzius/Applied-Data-Science-Capstone/blob/main/jupyter-labs-spacex-data-collection-api.ipynb>



Data Collection - Scraping

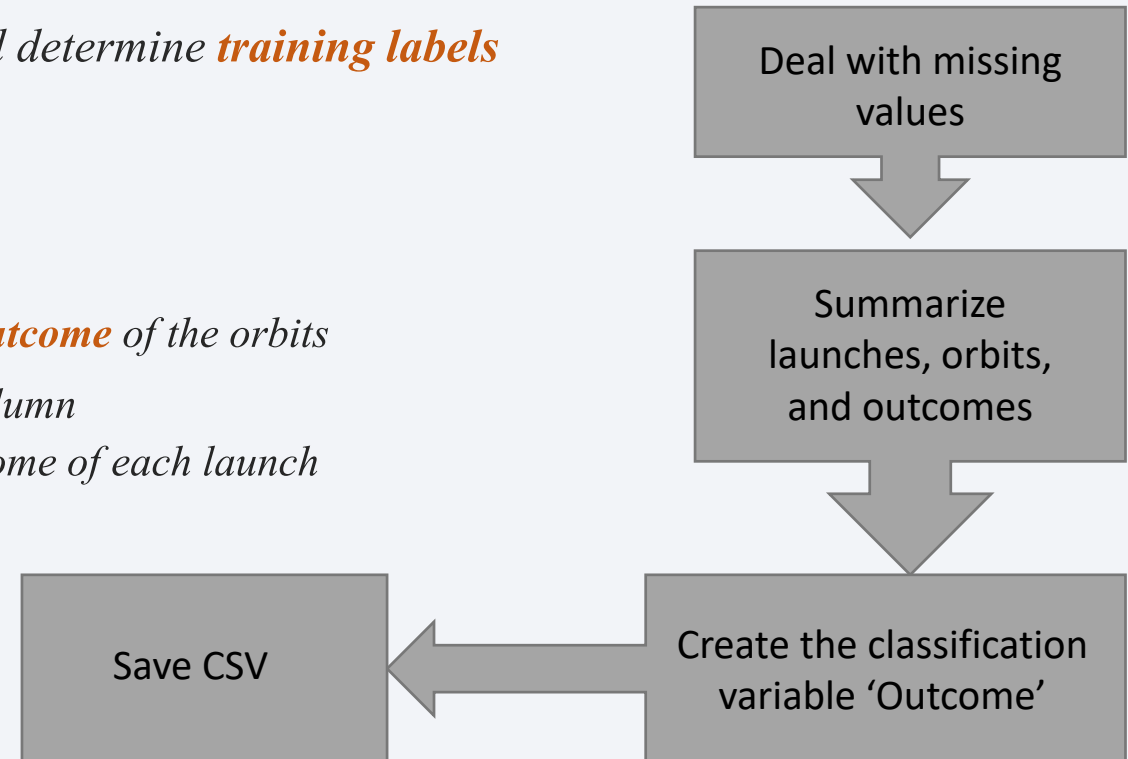
- *Web scrap Falcon 9 launch wikipage records with BeautifulSoup: extract the launch records HTML tables from Wikipedia, parse them and convert it into a Pandas data frame, save as CSV file*
- <https://github.com/yuretzius/Applied-Data-Science-Capstone/blob/main/jupyter-labs-webscraping.ipynb>



Data Wrangling

- Perform **initial exploratory data analysis (EDA)** and determine **training labels**

- ❖ identify and quantify **missing values**
- ❖ calculate the **number of launches** on each site
- ❖ calculate the number and occurrence of each **orbit**
- ❖ calculate the number and occurrence of **mission outcome** of the orbits
- ❖ create a landing outcome label from 'Outcome' column
the **classification variable** that represents the outcome of each launch
- ❖ calculate the **success rate**
- ❖ save as CSV file



- <https://github.com/yuretzius/Applied-Data-Science-Capstone/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb>

EDA with Data Visualization

The following plots were created:

- **Flight Number vs. Payload Mass** category plot, with Outcome overlaid:
dependence of success on continuous launch attempts and payload
- **Flight Number vs. Launch Site** category plot, with Outcome overlaid:
dependence of success on launch site as launch attempts progressed
- **Payload Mass vs. Launch Site** category plot, with Outcome overlaid:
determine the relationship between launch sites and payloads, and see if it influences the success
- **Success of each Orbit Type** bar chart: *check if some orbits have better success rate than others*
- **Flight Number vs. Orbit Type** category plot, with Outcome overlaid:
dependence of success on orbit as launch attempts progressed
- **Payload Mass vs. Orbit Type** category plot, with Outcome overlaid:
see the relationship between payload mass and orbit, and how it is related to success
- **Success Rate yearly trend** line plot: *how launch success progresses with time*

In addition, additional features were engineered: **dummy variables** created for **categorical columns**

<https://github.com/yuretzius/Applied-Data-Science-Capstone/blob/main/edadataviz.ipynb>

EDA with SQL

The following SQL queries were performed:

- *names of the unique launch sites*
- *5 records where launch sites begin with the string 'CCA'*
- *total payload mass carried by boosters launched by NASA (CRS)*
- *average payload mass carried by booster version F9 v1.1*
- *date when the first successful landing outcome in ground pad was achieved*
- *names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000*
- *total number of successful and failure mission outcomes*
- *all the booster versions that have carried the maximum payload mass*
- *failure landing outcomes in drone ship, booster versions, and launch sites for the months in year 2015*
- *rank the count of landing outcomes between the dates 2010-06-04 and 2017-03-20*
- https://github.com/yuretzius/Applied-Data-Science-Capstone/blob/main/jupyter-labs-eda-sql-coursera_sqlite.ipynb

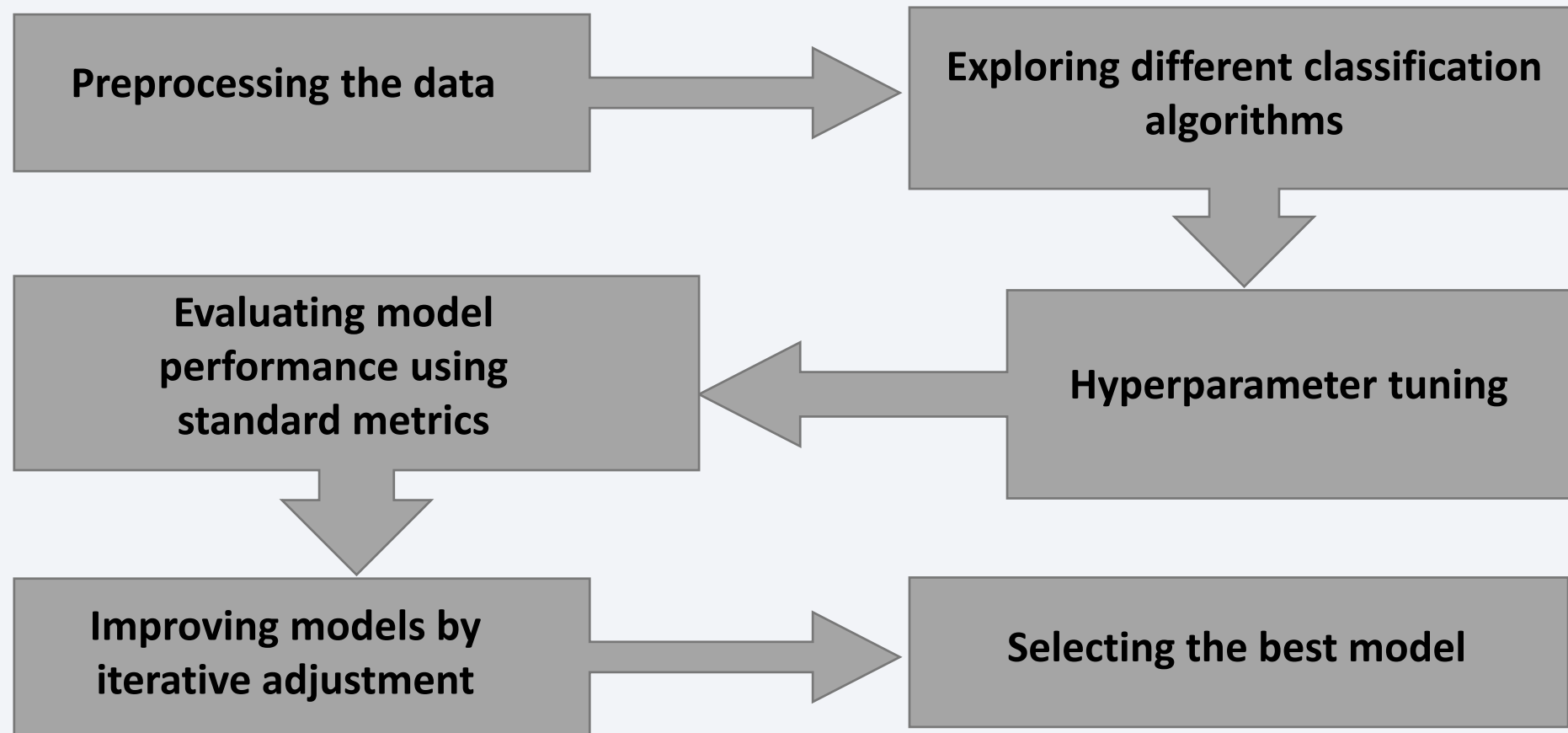
Build an Interactive Map with Folium

- *Circle: to mark NASA Johnson Space Center at Houston, Texas*
- *Circles and marks: for each of the launch sites.*
- *Marker clusters: to indicate successful and failed launches at each of the sites*
- *Line and marker: to show the shortest distance from one of the launch sites to the coastline*
- *Line and marker: to the city center of the town of Titusville*
- *Reasons: to provide visual aids for the data on launches and proximity to the important landmarks.*
- https://github.com/yuretzius/Applied-Data-Science-Capstone/blob/main/lab_jupyter_launch_site_location.ipynb

Build a Dashboard with Plotly Dash

- *Plots / Graphs:*
 - ❖ *pie chart* of success count for launch sites
 - ❖ *scatter plot* of payload mass vs. success / failure with booster type overlaid
- *Interactions:*
 - ❖ *dropdown switch* between all launch sites and each of them in particular
 - ❖ *payload mass range slider*
- *Reasons:*
 - ❖ to *show the relationship* between site location, success / failure rate, payload mass, and booster type.
- <https://github.com/yuretzius/Applied-Data-Science-Capstone/blob/main/spacex-dash-app.py>

Predictive Analysis (Flowchart)



Predictive Analysis (Overview)

- **Preprocessed** the data, by **standardizing** features and **splitting** it into training and test sets.
- Explored various **classification algorithms**, by trying **logistic regression**, **support vector machine**, **decision tree**, and **K-nearest neighbor**.
- Performed **hyperparameter tuning** on each of them: used **GridSearchCV** to **fine tune hyperparameters** C for logreg, kernel, C , and gamma for SVM, six hyperparameters like max depth and max features for decision tree, and n neighbors, algorithm, and p for KNN.
- Evaluated each **model performance**, using **confusion matrices** and calculating metrics like **accuracy**, **precision**, **recall**, **f1-score**, and **support**.
- Adjusted the models by doing **improvement iterations** based on the evaluation results.
- Selected **the best performing model**.
- https://github.com/yuretzius/Applied-Data-Science-Capstone/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb

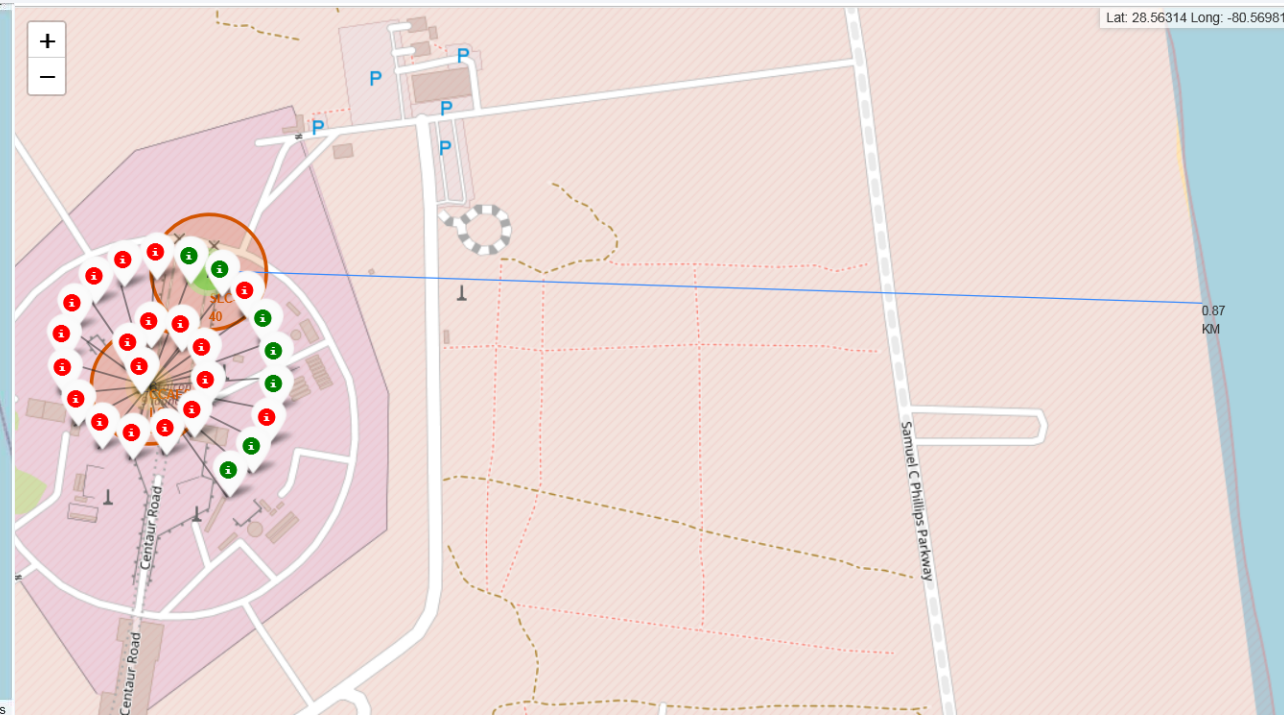
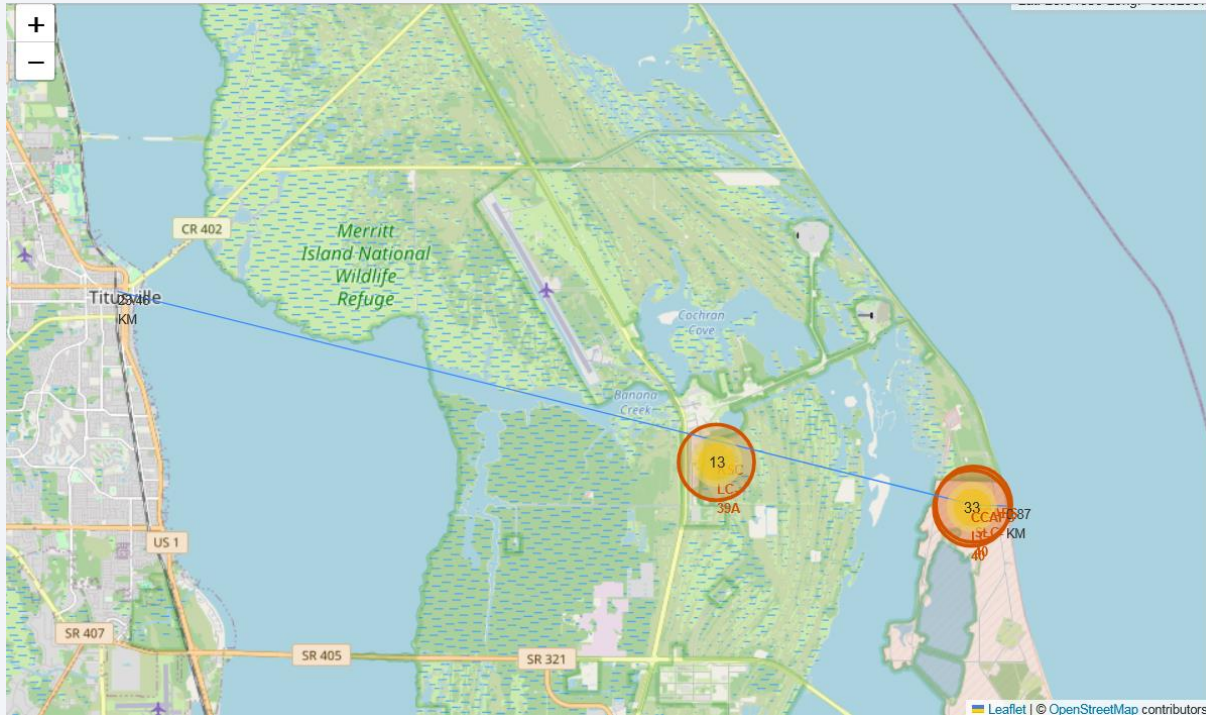
Results: Exploratory Data Analysis Results (Visualizations)

- *Most launches were made from CCAFS-SLC-40 launch site, with failures mostly for low payloads.*
- *For the VAFB-SLC-4E launch site there are no rockets launched for heavy payload mass (>10000).*
- *ES-L1, GEO, HEO, and SSO orbits have the highest success rate of 100%, while GTO has the lowest of $\sim 50\%$.*
- *With heavy payloads the successful landing or positive landing rate are more for Polar, LEO, and ISS.*
- *However, for GTO, it's difficult to distinguish between successful and unsuccessful landings as both outcomes are present.*
- *With heavy payloads the successful landing or positive landing rate are more for Polar, LEO, and ISS. However, for GTO, it's difficult to distinguish between successful and unsuccessful landings as both outcomes are present.*
- *The success rate since 2013 kept increasing till 2020.*

Results: Exploratory Data Analysis Results (SQL)

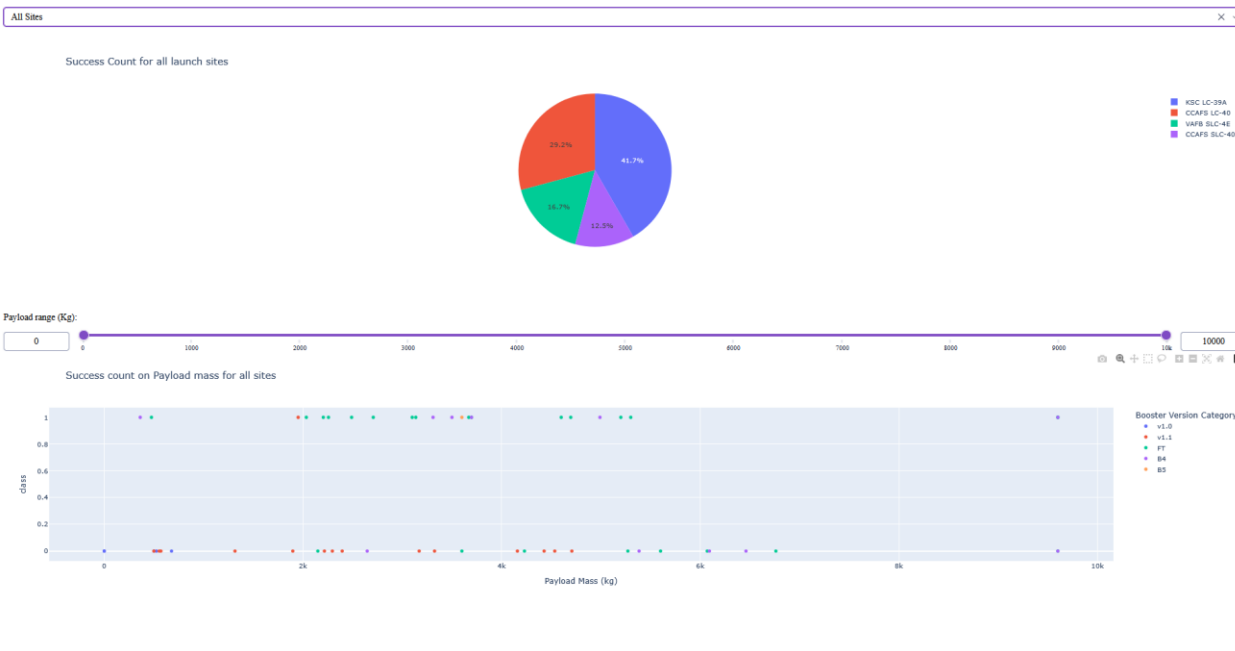
- *Unique launch sites were CCAFS-SLC-40, VAFB-SLC-4E, and KSC-LC-39A*
- *The total payload mass carried by boosters launched by NASA (CRS) was 45,596 kg.*
- *The average payload mass carried by booster version F9 v1.1 was 2928.4 kg.*
- *The date when the first successful landing outcome in ground pad was achieved was December 22, 2015.*
- *The names of the boosters which have success in drone ship and have payload mass >4000kg, but <6000 kg: F9 FT B1022, F9 FT B1026, F9 FT B1021.2, and F9 FT B1031.2.*
- *The total number of mission outcomes were 99 success, 1 success with unclear payload status, and 1 failure.*
- *All the booster versions that have carried the maximum payload mass of 15,600kg were F9 B5: B1048.4, B1049.4, B1051.3, B1056.4, B1048.5, B1051.4, B1049.5, B1060.2, B1058.3, B1051.6, B1060.3, and B1049.7*
- *Failure landings onto a drone ship in 2015 were F9 v1.1 B1012 with payload SpaceX CRS-5 in January and F9 v1.1 B1015 with payload SpaceX CRS-6, both launched from CCAFS-SLC-40.*
- *Landing outcomes between 2010-06-04 and 2017-03-20 were 10 unattempted, 5 successes on drone ship, 5 failures on drone ship, 3 successes on ground pad, 3 controlled ocean descents, 2 uncontrolled ocean descents, 2 parachute failures, and 1 precluded on a drone ship.*

Results: Interactive Analytics Demo in Screenshots (Folium)

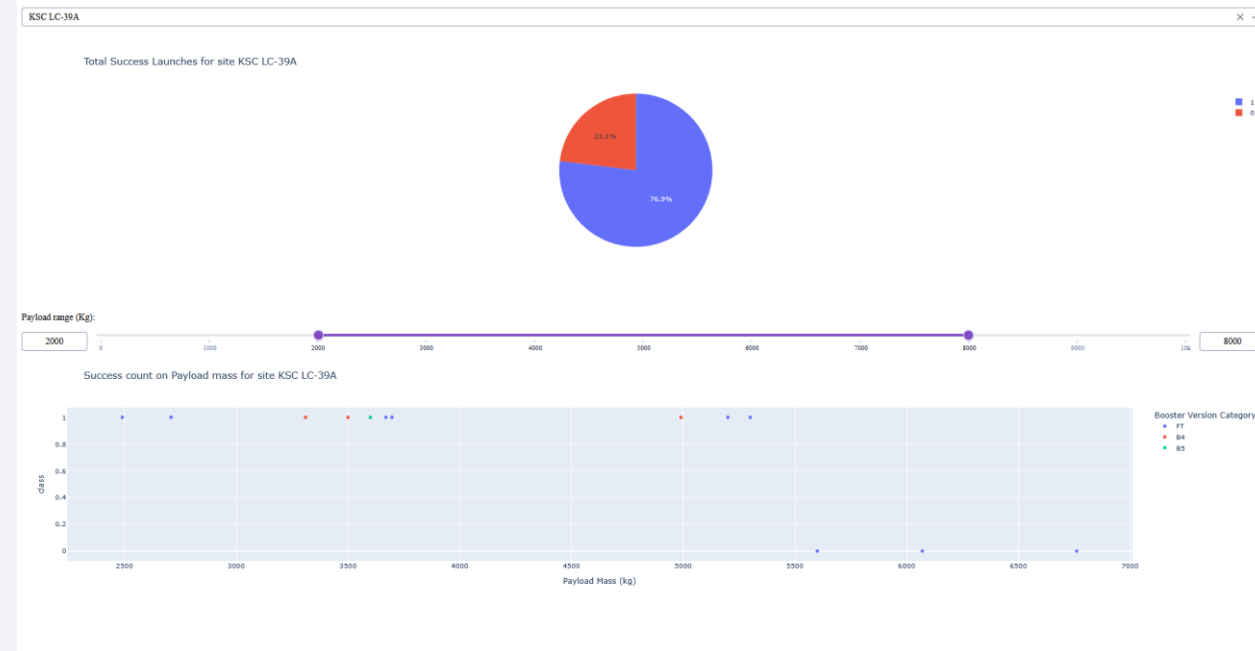


Results: Interactive Analytics Demo in Screenshots (Plotly Dash)

SpaceX Launch Records Dashboard



SpaceX Launch Records Dashboard



Results: Predictive Analysis Results

Logistic Regression:

	precision	recall	f1-score	support
0	1.00	0.50	0.67	6
1	0.80	1.00	0.89	12
accuracy			0.83	18
macro avg	0.90	0.75	0.78	18
weighted avg	0.87	0.83	0.81	18

Decision Tree:

	precision	recall	f1-score	support
0	1.00	0.83	0.91	6
1	0.92	1.00	0.96	12
accuracy			0.94	18
macro avg	0.96	0.92	0.93	18
weighted avg	0.95	0.94	0.94	18

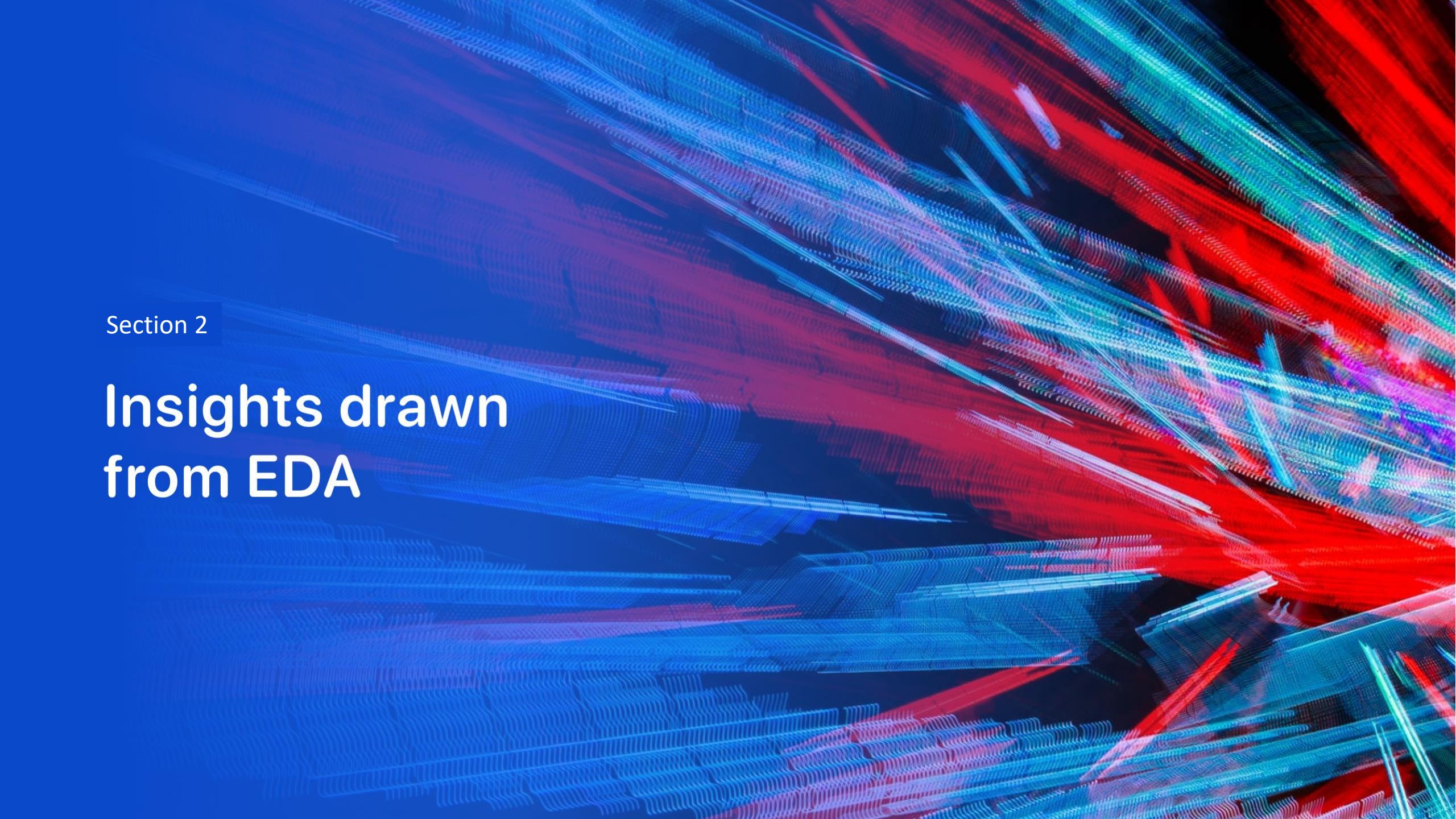
SVM:

	precision	recall	f1-score	support
0	1.00	0.50	0.67	6
1	0.80	1.00	0.89	12
accuracy			0.83	18
macro avg	0.90	0.75	0.78	18
weighted avg	0.87	0.83	0.81	18

kNN:

	precision	recall	f1-score	support
0	1.00	0.50	0.67	6
1	0.80	1.00	0.89	12
accuracy			0.83	18
macro avg	0.90	0.75	0.78	18
weighted avg	0.87	0.83	0.81	18

- **Decision Tree Classifier** performs better than the other three methods, but **needs a few reruns to get the good result**

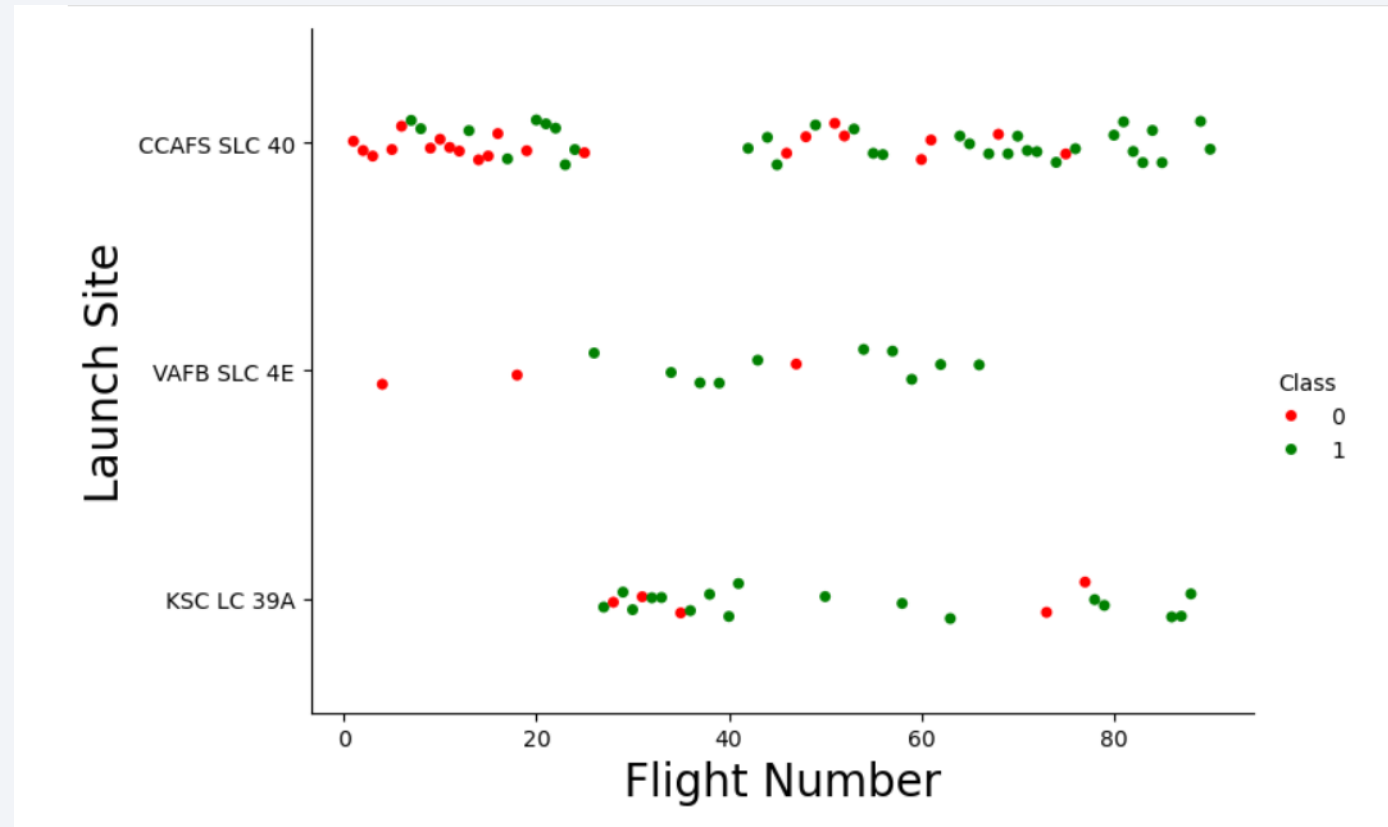
The background of the slide is an abstract composition. It features a dark blue field on the left side, which transitions into a complex pattern of diagonal streaks in shades of blue, red, and cyan on the right. These streaks have a textured, almost woven appearance. Overlaid on this pattern is a faint, light blue grid that recedes into the distance, creating a sense of depth and perspective.

Section 2

Insights drawn from EDA

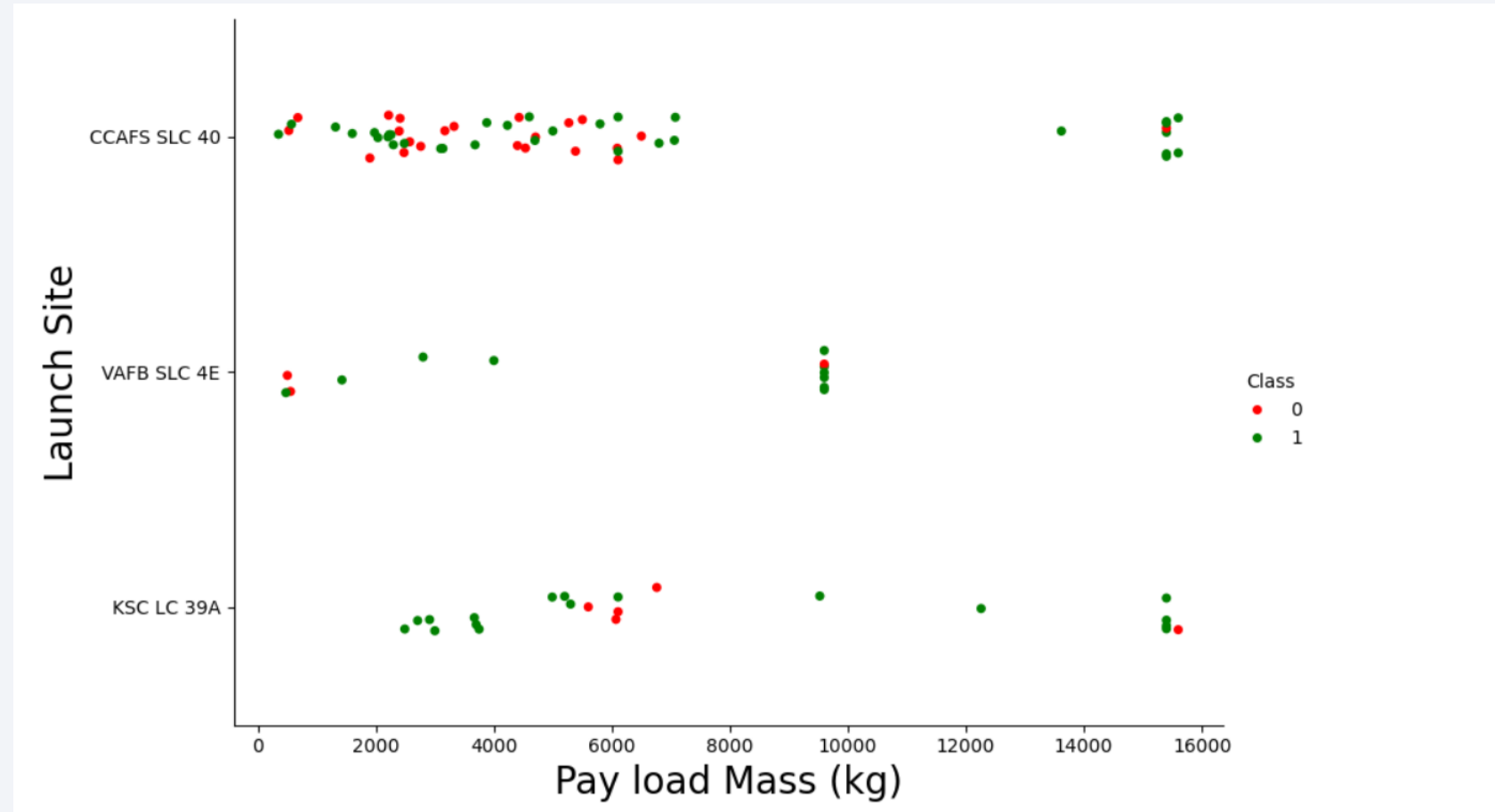
Flight Number vs. Launch Site

- *There is no equal distribution, so launch site is an important feature, influencing the flight outcome.*
- *The earliest launches had most failures, so flight number is also an important feature.*
- *Most launches were made from CCAFS-SLC-40.*
- *Of the three sites, KSC-LC-39A was the latecomer, launches started there only after more than 20 flights.*
- *VAFB-SLC-4E site was not used after about 65 launches.*



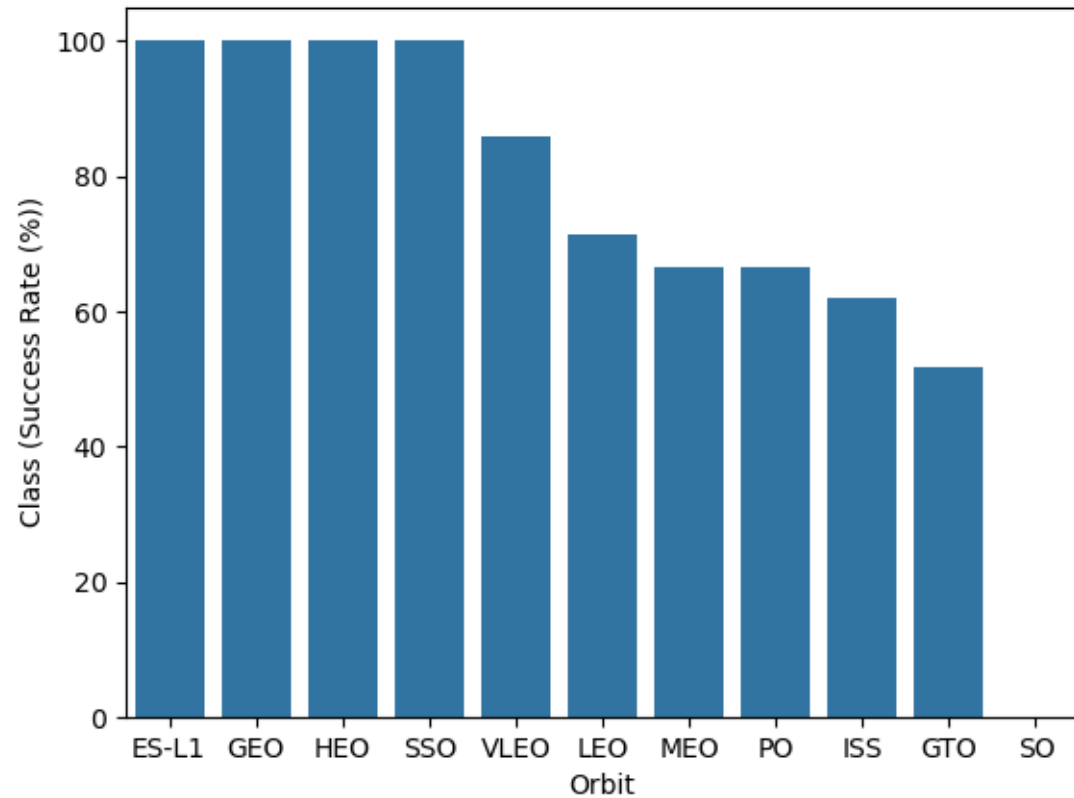
Payload vs. Launch Site

- *VAFB-SLC-4E is used for lighter payloads, while KSC-LC-39A handles heavier ones.*
- *CCAPS-SLC-40 is used for both a lighter range and several very heavy launches, but not in the payload midrange.*
- *Most failures correspond to light payloads launched from CCAPS-SLC-40 site.*



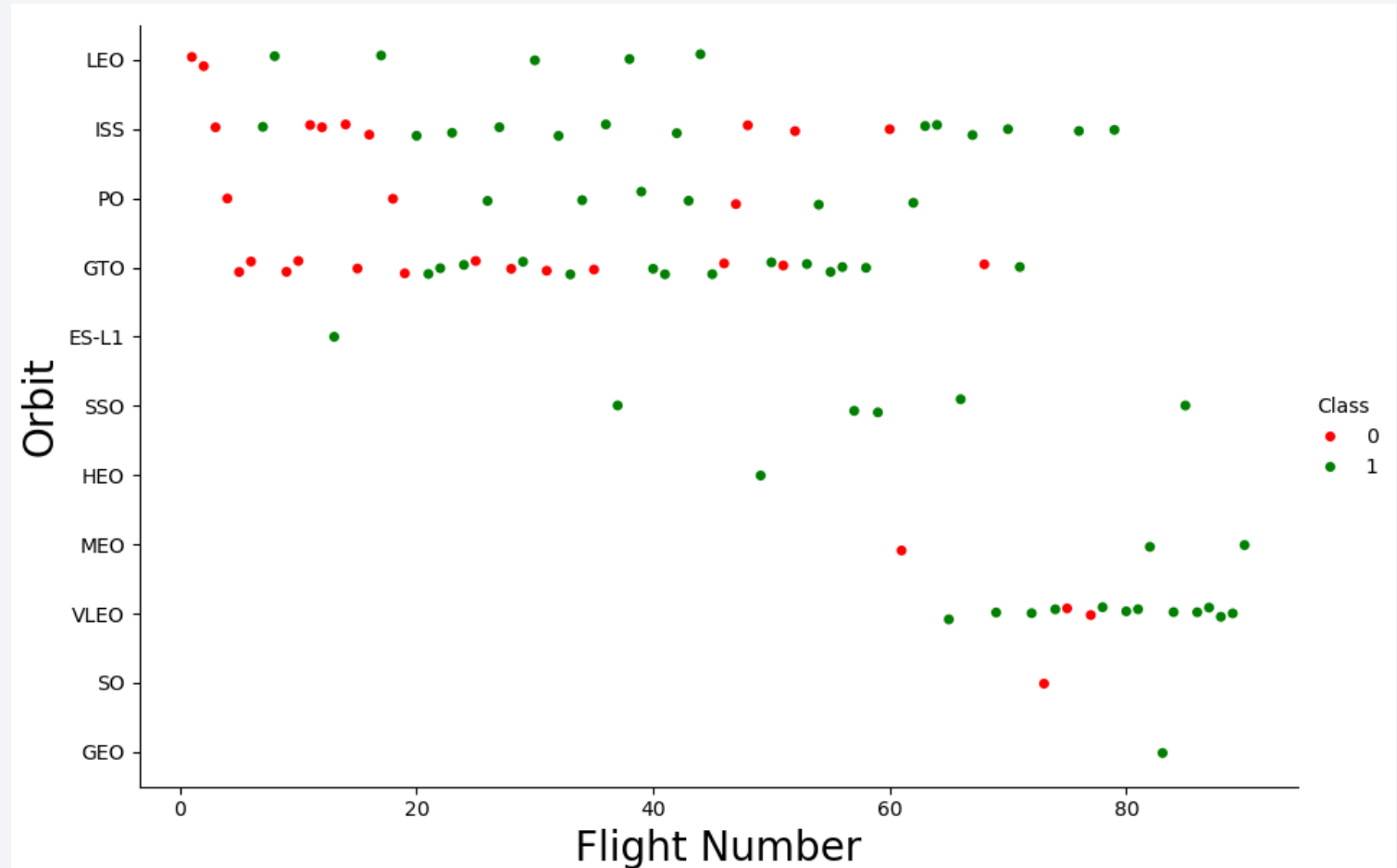
Success Rate vs. Orbit Type

- *ES-L1, GEO, HEO, and SSO orbits have the highest success rate of 100%.*
- *GTO orbit has the lowest success rate of ~50%.*
- *The rest of the orbits, VLEO, LEO, MEO, PO, and ISS occupy the middle ground with regard to success.*



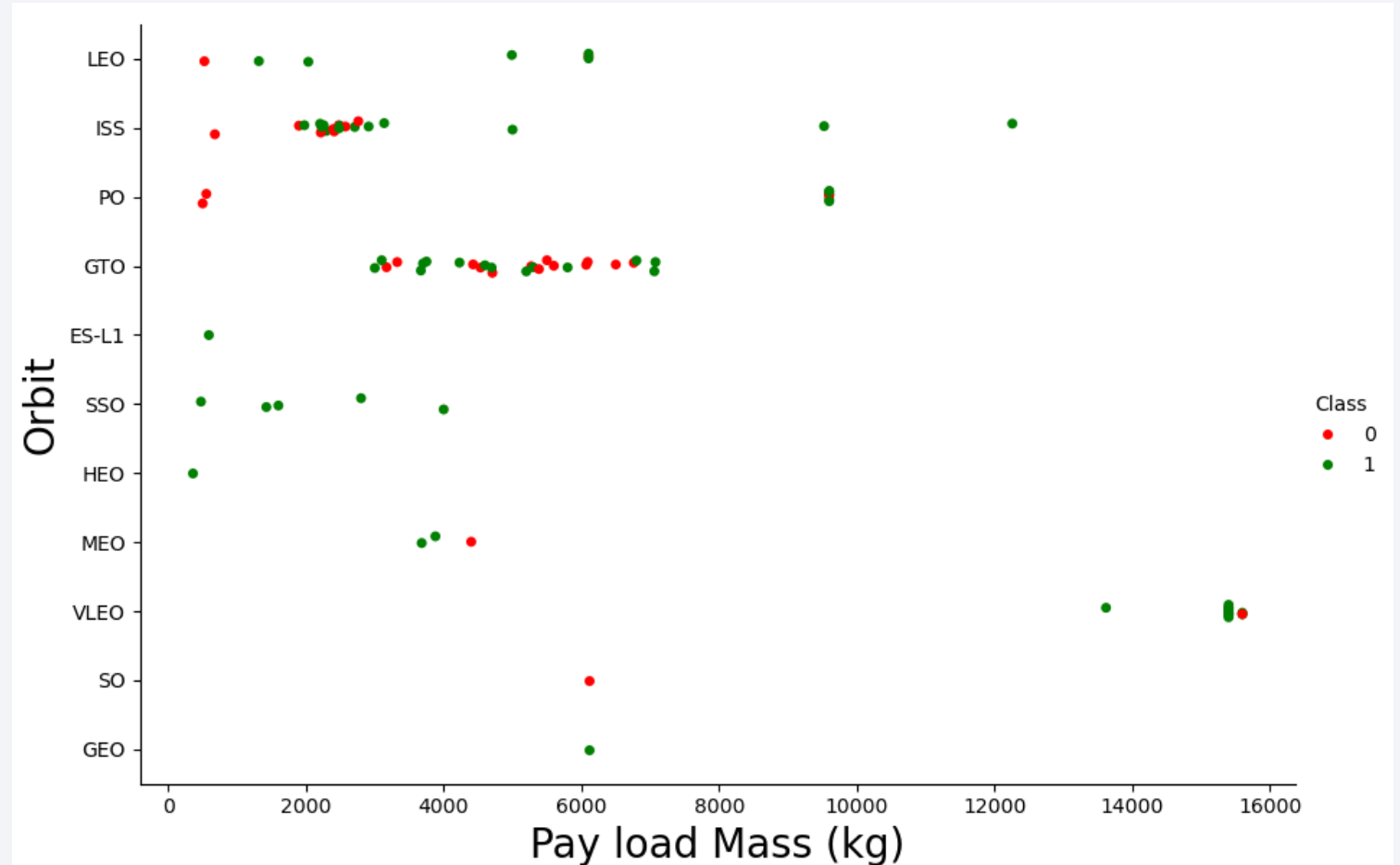
Flight Number vs. Orbit Type

- *Some orbits, like SSO, HEO, MEO, and VLEO, were only used later in time.*
- *On the contrary, LEO orbit was only used for earlier flights.*
- *ISS and GTO are the most continuously used orbits.*
- *Success distribution in ISS and GEO orbits is complex, one cannot say that only earlier flights led to failure.*



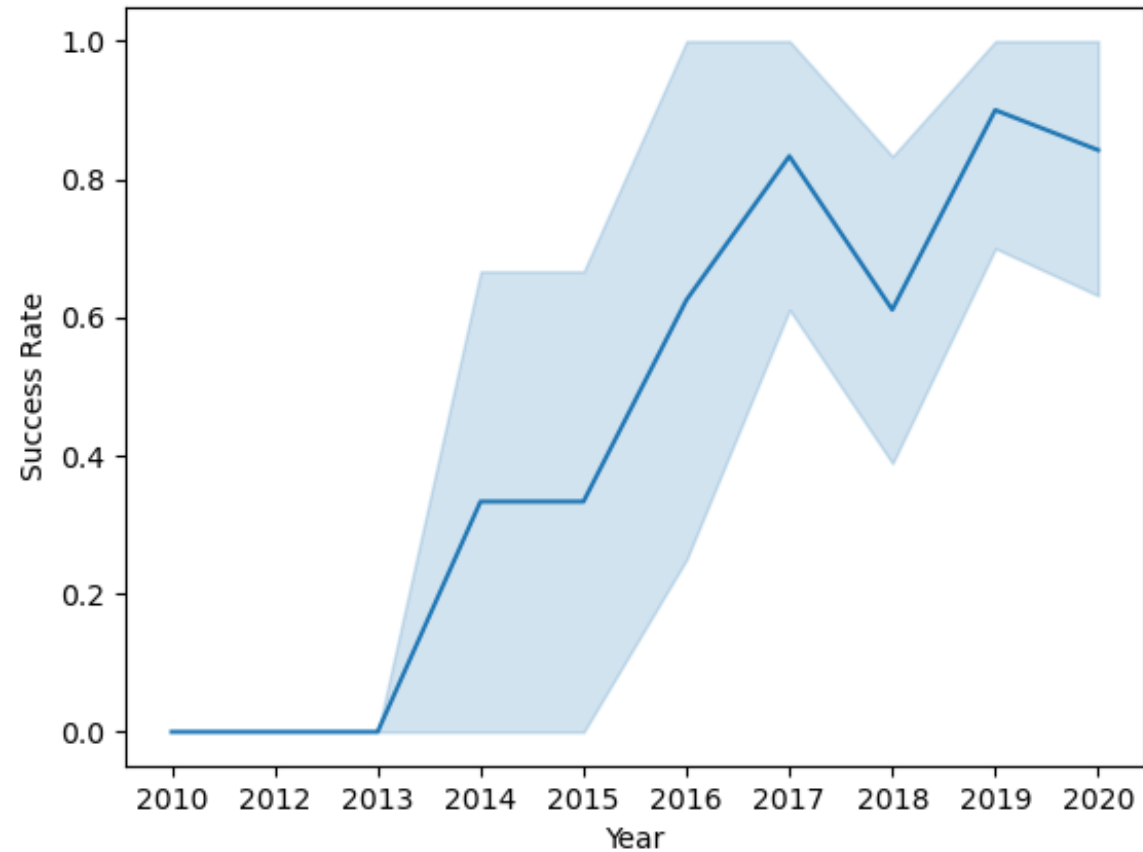
Payload vs. Orbit Type

- In many cases the orbit type determines the range of payload mass. E.g. GTO orbits corresponds to the narrow range of 3000 to 7000kg.
- Light payload orbits are LEO, ES-L1, SSO, HEO, and MEO.
- VLEO is a superheavy payload orbit.
- ISS covers the whole range of orbits, but most are clustered in the range of 2000-3000kg.
- PO orbit only had extra light <1000kg and moderately heavy ~10,000kg payloads.



Launch Success Yearly Trend

- *The success rate was steadily growing from 2013 to 2017, and then stayed about the same from 2017 to 2020.*
- *The slight fall in 2018 is probably a statistical fluctuation.*



All Launch Site Names

Task 1

Display the names of the unique launch sites in the space mission

```
[12]: %sql SELECT DISTINCT LAUNCH_SITE as "Launch_Sites" FROM SPACEXTBL;
```

```
* sqlite:///my_data1.db
```

Done.

```
[12]: Launch_Sites
```

```
CCAFS LC-40
```

```
VAFB SLC-4E
```

```
KSC LC-39A
```

```
CCAFS SLC-40
```

- *Distinct launch sites are: CCAFS SLC-40, CCAFS LC-40, VAFB SLC-4E, KSC LC-39A.*
- *CCAFS LC-40 and CCAFS SLC-40 are distinct, but almost the same (see the Folium maps for illustration) and for all intents and purposes should be treated as one.*

Launch Site Names Begin with 'CCA'

Task 2

Display 5 records where launch sites begin with the string 'CCA'

```
[13]: %sql SELECT * FROM 'SPACEXTBL' WHERE Launch_Site LIKE 'CCA%' LIMIT 5;
```

```
* sqlite:///my_data1.db  
Done.
```

```
[13]:
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

- *All five records correspond to CCAFS LC-40 launch site.*

Total Payload Mass

Task 3

Display the total payload mass carried by boosters launched by NASA (CRS)

```
[14]: %sql SELECT SUM(PAYLOAD_MASS__KG_) as "Total Payload Mass(Kgs)", Customer FROM 'SPACEXTBL' WHERE Customer = 'NASA (CRS)';  
* sqlite:///my_data1.db  
Done.
```

```
[14]: Total Payload Mass(Kgs)  Customer  
-----  
45596  NASA (CRS)
```

- *The total mass carried by boosters launched by NASA (CRS) is 45,596kg.*

Average Payload Mass by F9 v1.1

Task 4

Display average payload mass carried by booster version F9 v1.1

```
[15]: %sql SELECT AVG(PAYLOAD_MASS__KG_) as "Payload Mass Kgs", Customer, Booster_Version FROM 'SPACEXTBL' WHERE Booster_Version LIKE 'F9 v1.1';  
* sqlite:///my_data1.db  
Done.
```

```
[15]: Payload Mass Kgs Customer Booster_Version  
-----  
2928.4          SES          F9 v1.1
```

- *The average payload mass carried by booster version F9 v1.1 is 2928.4 kg.*

First Successful Ground Landing Date

Task 5

List the date when the first succesful landing outcome in ground pad was acheived.

Hint: Use min function

```
[20]: %sql SELECT MIN(DATE) FROM 'SPACEXTBL' WHERE "Landing_Outcome" = "Success (ground pad)";
```

```
* sqlite:///my_data1.db
```

Done.

```
[20]: MIN(DATE)
```

```
2015-12-22
```

- *The date of the first successful landing on a ground pad is 22 December 2015.*

Successful Drone Ship Landing with Payload between 4000 and 6000

Task 6

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

```
[23]: %sql SELECT DISTINCT Booster_Version, Payload FROM SPACEXTBL WHERE "Landing_Outcome" = "Success (drone ship)" AND PAYLOAD_MASS_KG_ < 6000 AND PAYLOAD_MASS_KG_ > 4000
```

```
* sqlite:///my_data1.db  
Done.
```

```
[23]:
```

Booster_Version	Payload
F9 FT B1022	JCSAT-14
F9 FT B1026	JCSAT-16
F9 FT B1021.2	SES-10
F9 FT B1031.2	SES-11 / EchoStar 105

- *The full SQL query is:*
`SELECT DISTINCT Booster_Version, Payload FROM SPACEXTBL WHERE "Landing_Outcome" = "Success (drone ship)" AND PAYLOAD_MASS_KG_ < 6000 AND PAYLOAD_MASS_KG_ > 4000;`
- *The names of boosters which have successfully landed on drone ship and had payload mass between 4000 and 6000 kg were **F9 FT B1022**, **F9 FT B1026**, **F9 FT B1021.2**, and **F9 FT B1031.2**.*

Total Number of Successful and Failure Mission Outcomes

Task 7

List the total number of successful and failure mission outcomes

```
[24]: %sql SELECT "Mission_Outcome", COUNT("Mission_Outcome") as Total FROM SPACEXTBL GROUP BY "Mission_Outcome";  
* sqlite:///my_data1.db  
Done.
```

```
[24]:
```

Mission_Outcome	Total
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

- The total number of mission outcomes were 99 success, 1 success with unclear payload status, and 1 failure.

Boosters Carried Maximum Payload

Task 8

List all the booster_versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function.

```
[25]: %sql SELECT "Booster_Version",Payload, "PAYLOAD_MASS__KG_" FROM SPACEXTBL WHERE "PAYLOAD_MASS__KG_" = (SELECT MAX("PAYLOAD_MASS__KG_") FROM SPACEXTBL)
* sqlite:///my_data1.db
Done.
```

[25]:	Booster_Version	Payload	PAYLOAD_MASS__KG_
	F9 B5 B1048.4	Starlink 1 v1.0, SpaceX CRS-19	15600
	F9 B5 B1049.4	Starlink 2 v1.0, Crew Dragon in-flight abort test	15600
	F9 B5 B1051.3	Starlink 3 v1.0, Starlink 4 v1.0	15600
	F9 B5 B1056.4	Starlink 4 v1.0, SpaceX CRS-20	15600
	F9 B5 B1048.5	Starlink 5 v1.0, Starlink 6 v1.0	15600
	F9 B5 B1051.4	Starlink 6 v1.0, Crew Dragon Demo-2	15600
	F9 B5 B1049.5	Starlink 7 v1.0, Starlink 8 v1.0	15600
	F9 B5 B1060.2	Starlink 11 v1.0, Starlink 12 v1.0	15600
	F9 B5 B1058.3	Starlink 12 v1.0, Starlink 13 v1.0	15600
	F9 B5 B1051.6	Starlink 13 v1.0, Starlink 14 v1.0	15600
	F9 B5 B1060.3	Starlink 14 v1.0, GPS III-04	15600
	F9 B5 B1049.7	Starlink 15 v1.0, SpaceX CRS-21	15600

- *Full SQL query:*

```
SELECT "Booster_Version",Payload, "PAYLOAD_MASS__KG_"  
FROM SPACEXTBL WHERE "PAYLOAD_MASS__KG_" = (SELECT  
MAX("PAYLOAD_MASS__KG_") FROM SPACEXTBL);
```
- *All the booster versions that have carried the maximum payload mass of 15,600kg were F9 B5: B1048.4, B1049.4, B1051.3, B1056.4, B1048.5, B1051.4, B1049.5, B1060.2, B1058.3, B1051.6, B1060.3, and B1049.7*

2015 Launch Records

Task 9

List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.

Note: SQLite does not support monthnames. So you need to use substr(Date, 6,2) as month to get the months and substr(Date,0,5)='2015' for year.

```
[29]: %sql SELECT substr(Date,6,2) as "Month", substr(Date,0,5) as "Year", "Booster_Version", "Launch_Site", Payload, "PAYLOAD_MASS_KG_", "Mission_Outcome"
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
[29]:
```

Month	Year	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Mission_Outcome	Landing_Outcome
01	2015	F9 v1.1 B1012	CCAFS LC-40	SpaceX CRS-5	2395	Success	Failure (drone ship)
04	2015	F9 v1.1 B1015	CCAFS LC-40	SpaceX CRS-6	1898	Success	Failure (drone ship)

- *Full SQL query:*

```
SELECT substr(Date,6,2) as "Month", substr(Date,0,5) as "Year",  
       "Booster_Version", "Launch_Site", Payload, "PAYLOAD_MASS_KG_", "Mission_Outcome", "Landing_Outcome"  
FROM SPACEXTBL WHERE substr(Date,0,5)='2015' AND "Landing_Outcome" = 'Failure (drone ship)';
```

- *Failure landings onto a drone ship in 2015 were F9 v1.1 B1012 with payload SpaceX CRS-5 in January and F9 v1.1 B1015 with payload SpaceX CRS-6, both launched from CCAFS-SLC-40.*

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Task 10

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

```
[35]: %sql SELECT "Landing_Outcome", COUNT("Landing_Outcome") as Total FROM SPACEXTBL WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY "Landing_Outcome"
```

```
* sqlite:///my_data1.db  
Done.
```

```
[35]:
```

Landing_Outcome	Total
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

- *Full SQL query:*

```
SELECT "Landing_Outcome", COUNT("Landing_Outcome") as Total FROM SPACEXTBL  
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'  
GROUP BY "Landing_Outcome" ORDER BY Total DESC;
```

- *Landing outcomes between 2010-06-04 and 2017-03-20 were 10 unattempted, 5 successes on drone ship, 5 failures on drone ship, 3 successes on ground pad, 3 controlled ocean descents, 2 uncontrolled ocean descents, 2 parachute failures, and 1 precluded on a drone ship*

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

Section 3

Launch Sites Proximities Analysis

All Launch Sites as Shown on an Interactive Folium Map



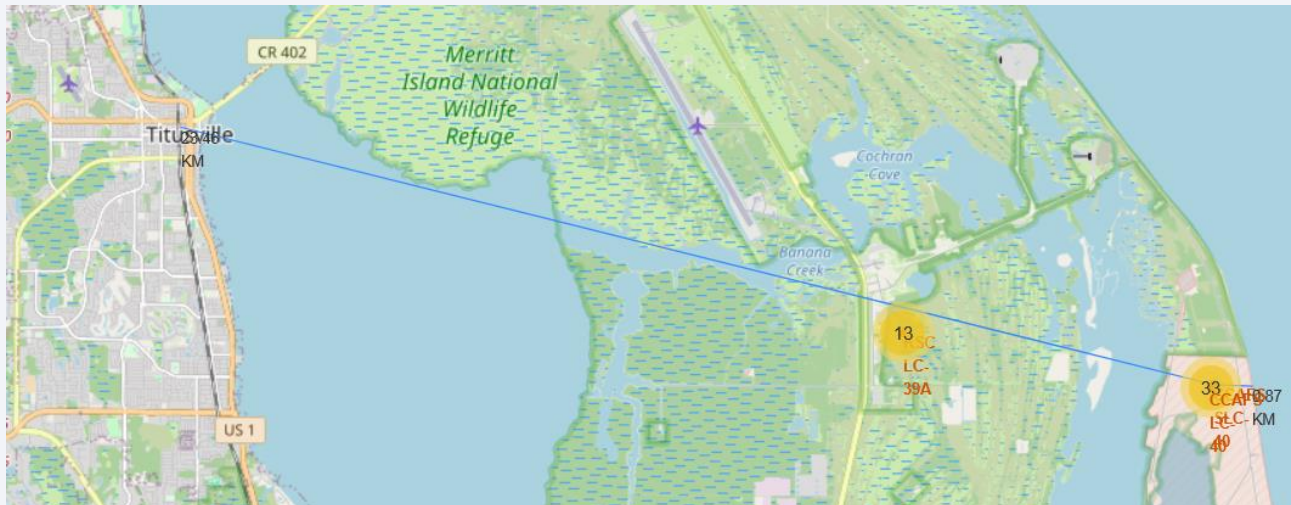
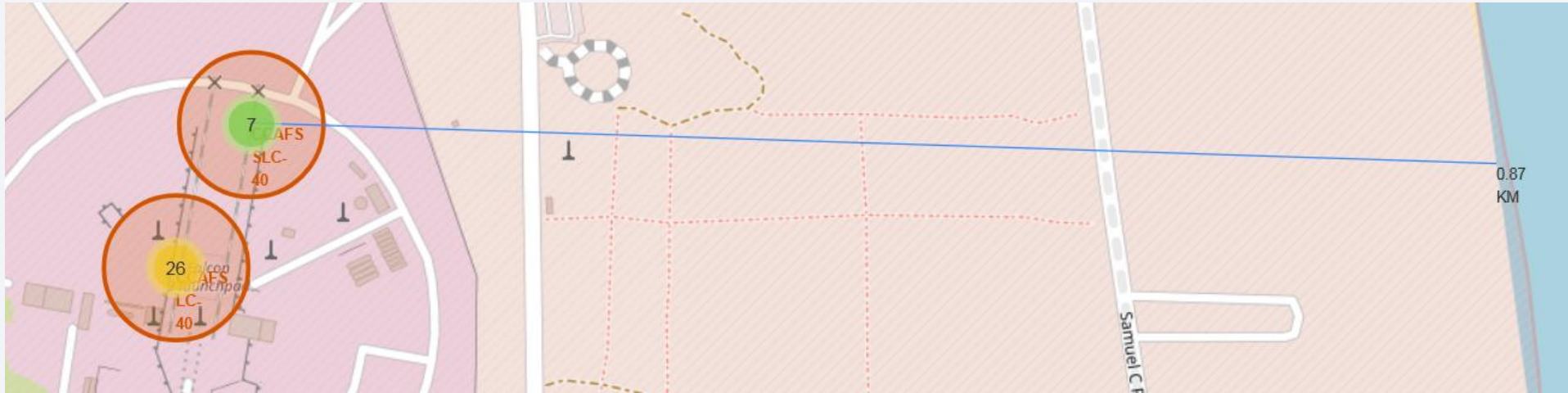
- All three sites are relatively close to the equator and very close to the coast.
- Two are on the east coast and one on the west coast.

The Color-labeled Launch Outcomes on the Folium Interactive Map

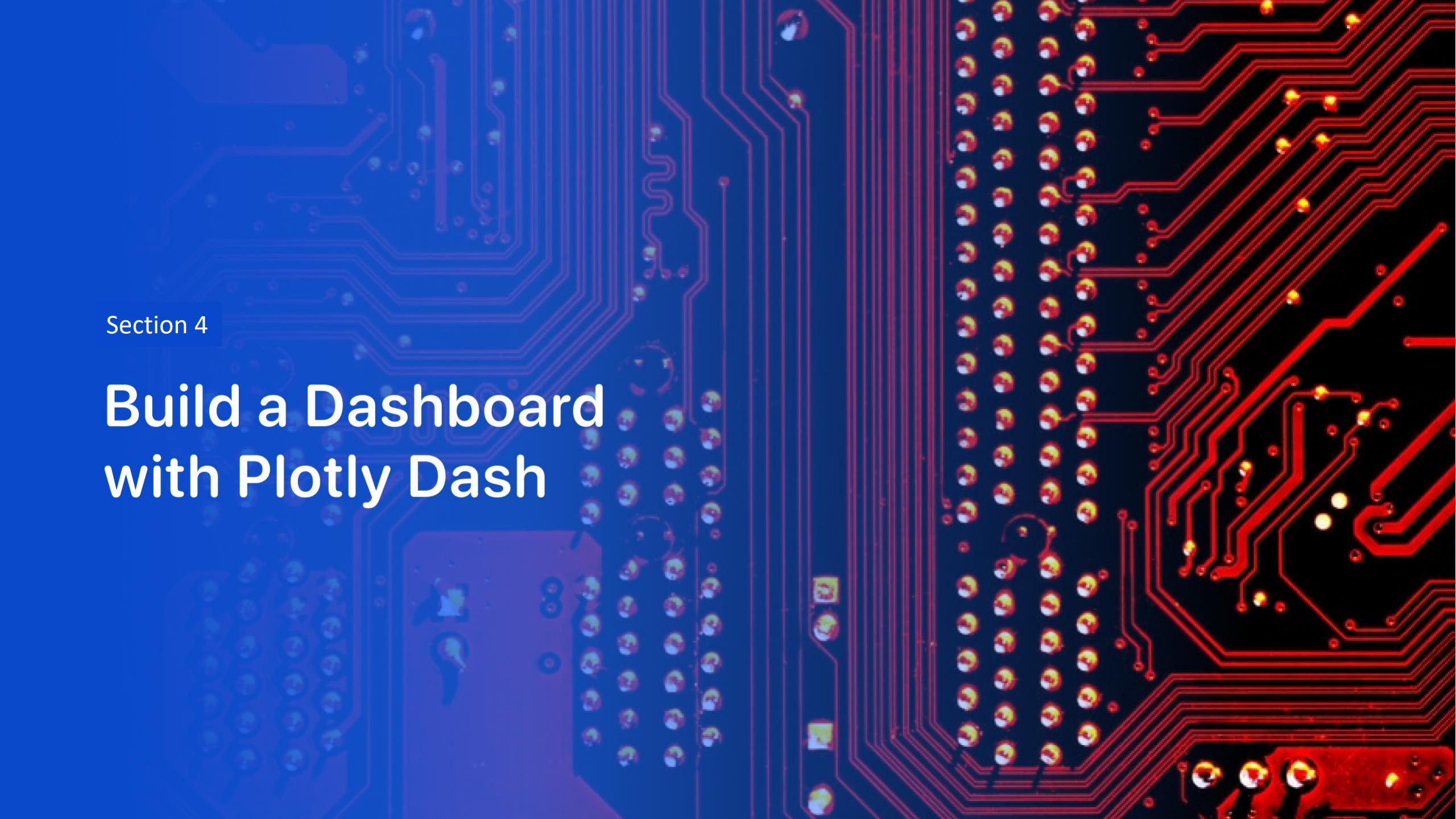


- *We can immediately see the distribution of successful and failed landing outcomes for all four launch sites, including the very close CCAFS LC-40 and CCAFS SLC-40.*
- *KSC LC-39A seems to be the most successful of all.*

Proximities to Important Landmarks on the Folium Interactive Map



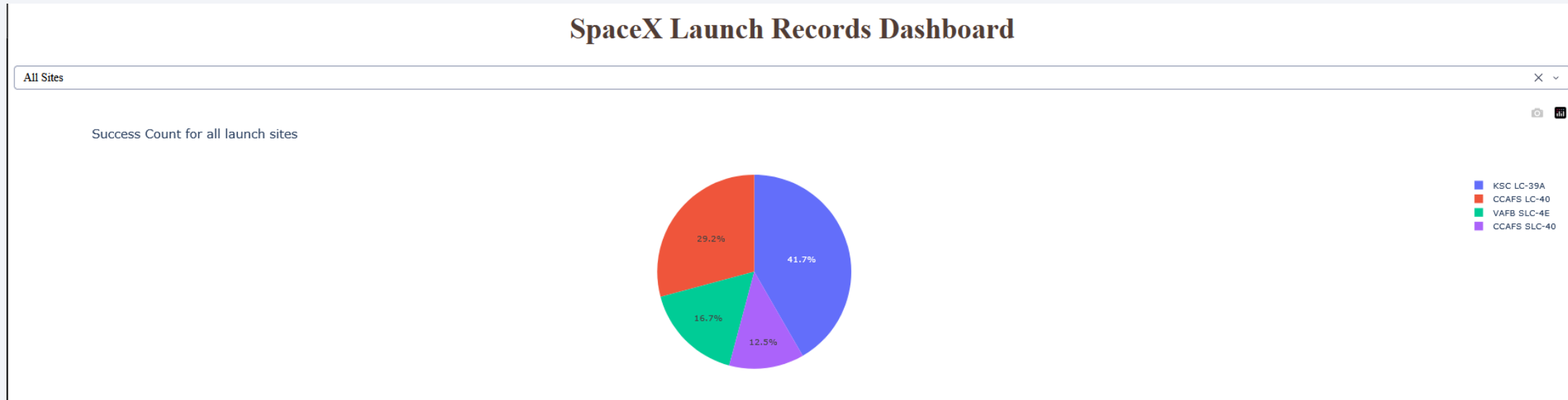
- *Distance from CCAFS LC-40 to the coastline is just ~870m.*
- *Distance from CCAFS LC-40 / SLC-40 to the center of the town of Titusville is ~23 km.*



Section 4

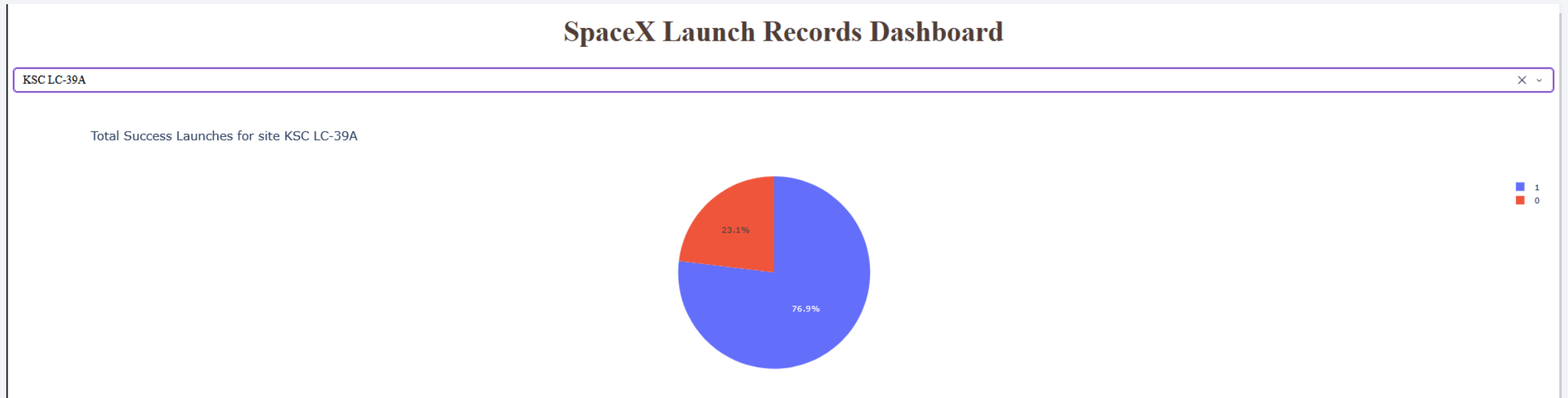
Build a Dashboard with Plotly Dash

Success Count for All Sites, in a Piechart



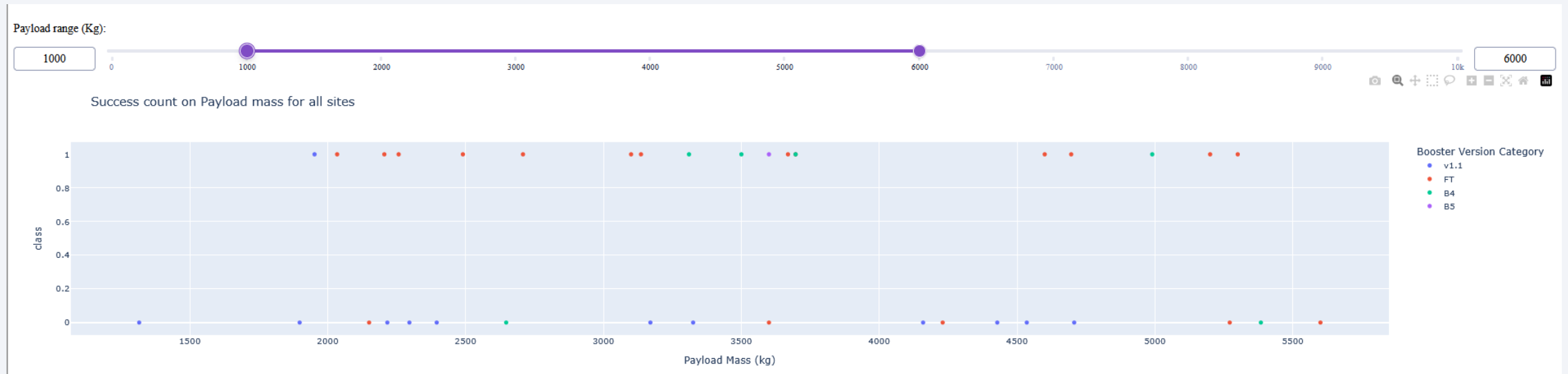
- *KSC LC-39A contributes the most of the success count of 41.7%, confirming the observation we did on the Folium map.*
- *CCAFS SLC-40 contributes the least, only 12.5%, but its close counterpart LC-40 adds 29.2%.*
- *The west coast VAFB SLC-4E adds 16.7%.*

Launch Site with the Best Success Ratio: KSC LC-39A



- The best performance with regard to the success ratio is displayed by KSC LC-39A, with 76.9% successes and only 23.1% failures*

Payload vs. Outcome scatter plot with Payload Range 1000-6000



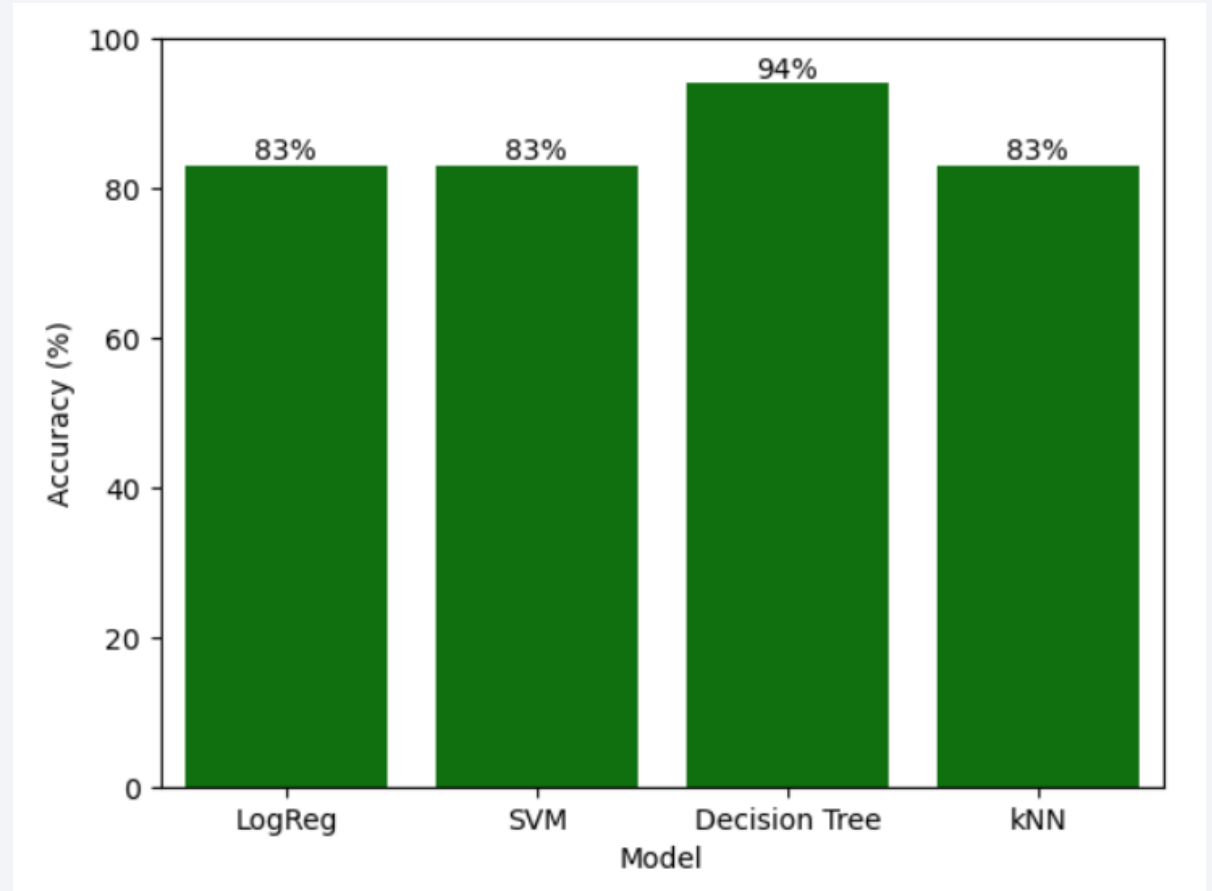
- *This plot shows the distribution of successes and failures of different booster versions in the lower range of payloads from 1000 to 6000 kg.*
- *The distribution does not show any obvious biases and is roughly homogeneous.*

Section 5

Predictive Analysis (Classification)

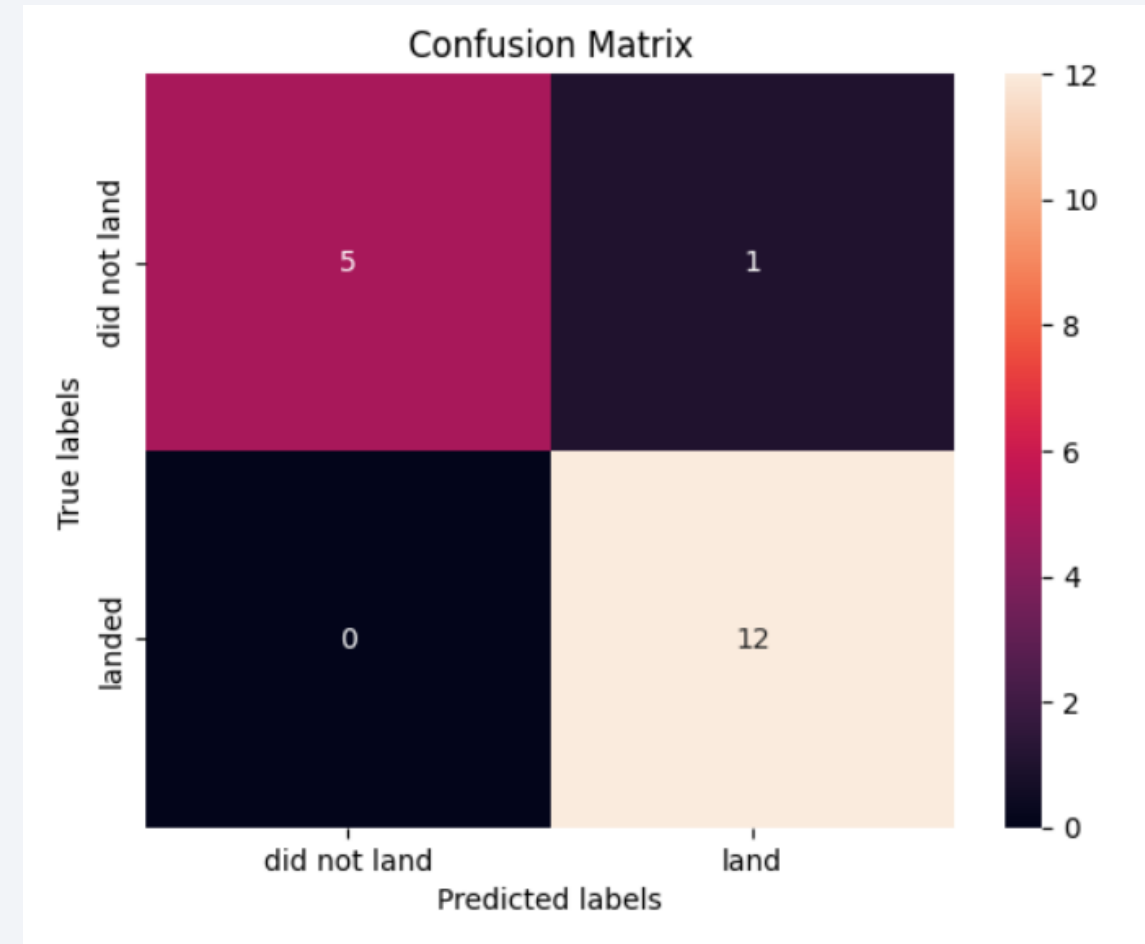
Classification Accuracy

- The best performing model is the **Decision Tree** classifier with 94% accuracy, achieved after a few reruns
- All the other three models: **logistic regression**, **support vector machine**, and **k-nearest neighbor** show the same accuracy of 83%



Confusion Matrix

- *The best performing model is the Decision Tree classifier*
- *It shows near perfect performance with only one false positive and no false negatives.*
- *Out of 18 test data samples 17 were classified correctly*



Conclusions

POINT 1: *The best performance with regard to the success ratio is displayed by KSC LC-39A launch site with 76.9% successes. It is also responsible for 41.7% of the total successes. This indicates that this site may have optimal conditions for the effective launches.*

POINT 2: *ES-L1, GEO, HEO, and SSO orbits have the highest success rate of 100%, while the GTO orbits shown the worst performance of about 50%. It shows that the type of orbit is definitely a factor in the overall success.*

POINT 3: *In general there is a weak trend of lower payloads performing worse, especially below 6000kg, but it may be related to the timeline dependence: earlier launches carried less payload.*

POINT 4: *There is a clear pattern of improvement over time from 2013 to 2017, with the success rate stabilizing afterwards.*

POINT 5: *Although all predictive models performed well, the Decision Tree was the best, with 94% accuracy. It performed so well on the test data, that out of 18 data points there were only 1 false positive and no false negatives.*

Appendix

- *Datasets (saved at different stages of the project):*

https://github.com/yuretzius/Applied-Data-Science-Capstone/blob/main/dataset_part_1.csv

https://github.com/yuretzius/Applied-Data-Science-Capstone/blob/main/dataset_part_2.csv

https://github.com/yuretzius/Applied-Data-Science-Capstone/blob/main/dataset_part_3.csv

- *Python scripts:*

Data Collection (API): <https://github.com/yuretzius/Applied-Data-Science-Capstone/blob/main/DataCollectionAPI.py>

Data Collection (Web): <https://github.com/yuretzius/Applied-Data-Science-Capstone/blob/main/DataCollectionWeb.py>

Data Wrangling: <https://github.com/yuretzius/Applied-Data-Science-Capstone/blob/main/DataWrangling.py>

Visualization EDA: https://github.com/yuretzius/Applied-Data-Science-Capstone/blob/main/Visualization_EDA.py

SQL EDA: https://github.com/yuretzius/Applied-Data-Science-Capstone/blob/main/SQL_EDA.py

Folium Map: <https://github.com/yuretzius/Applied-Data-Science-Capstone/blob/main/FoliumLaunchSites.py>

Plotly/Dash: <https://github.com/yuretzius/Applied-Data-Science-Capstone/blob/main/spacex-dash-app.py>

Predictive Analysis: <https://github.com/yuretzius/Applied-Data-Science-Capstone/blob/main/ML.py>

Appendix

- *SQL queries:*

```
SELECT DISTINCT LAUNCH_SITE as "Launch_Sites" FROM SPACEXTBL;

SELECT * FROM 'SPACEXTBL' WHERE Launch_Site LIKE 'CCA%' LIMIT 5;

SELECT SUM(PAYLOAD_MASS__KG_) as "Total Payload Mass(Kgs)", Customer FROM 'SPACEXTBL' WHERE Customer = 'NASA (CRS)';

SELECT AVG(PAYLOAD_MASS__KG_) as "Payload Mass Kgs", Customer, Booster_Version FROM 'SPACEXTBL'
WHERE Booster_Version LIKE 'F9 v1.1';

SELECT MIN(DATE) FROM 'SPACEXTBL' WHERE "Landing_Outcome" = "Success (ground pad)";

SELECT DISTINCT "Landing_Outcome" FROM SPACEXTBL;

SELECT DISTINCT Booster_Version, Payload FROM SPACEXTBL
WHERE "Landing_Outcome" = "Success (drone ship)" AND PAYLOAD_MASS__KG_ < 6000 AND PAYLOAD_MASS__KG_ > 4000;

SELECT "Mission_Outcome", COUNT("Mission_Outcome") as Total FROM SPACEXTBL GROUP BY "Mission_Outcome";

SELECT "Booster_Version", Payload, "PAYLOAD_MASS__KG_" FROM SPACEXTBL
WHERE "PAYLOAD_MASS__KG_" = (SELECT MAX("PAYLOAD_MASS__KG_") FROM SPACEXTBL);

SELECT substr(Date,6,2) as "Month", substr(Date,0,5) as "Year",
"Booster_Version", "Launch_Site", Payload, "PAYLOAD_MASS__KG_", "Mission_Outcome", "Landing_Outcome"
FROM SPACEXTBL WHERE substr(Date,0,5)='2015' AND "Landing_Outcome" = 'Failure (drone ship)';

SELECT "Landing_Outcome", COUNT("Landing_Outcome") as Total FROM SPACEXTBL
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY "Landing_Outcome" ORDER BY Total DESC;
```


Thank you!

